

26 Reaction kinetics

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A rate law is determined experimentally: $\text{rate} = k[\text{A}]^m[\text{B}]^n$. Reaction order is not inferred automatically from balanced stoichiometry. A first-order reaction has constant half-life at fixed temperature. The Arrhenius equation links temperature, activation energy and rate constant; a catalyst lowers activation energy.

Worked example: follow the reasoning

If doubling [A] doubles rate while doubling [B] quadruples it, rate is first order in A and second in B.

Quick check · multiple choice

A When doubling [A] quadruples rate, order in A is what? [1]

- A First
- B Second
- C Zero
- D Fourth

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 When [A] doubles at fixed [B], rate rises fourfold; when [B] doubles, rate is unchanged. Write rate law and overall order. [3]

Answers and explanation

Quick check: B. Second

Question 1 [3 marks]

Order 2 in A [1], order 0 in B [1]; rate = $k[A]^2$, overall order 2 [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.